

Krishnaa Vadivel | Computational Physics | Distributed Computing | Machine Learning

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Summary

Computational physics researcher building distributed scientific software, machine-learning accelerators, and first-principles materials workflows. Experience spans Python/C++, CUDA/ROCm, MPI/OpenMP, Linux-based HPC environments, custom Docker containers for reproducibility, and scientific ML for excitonic materials problems.

Research Experience

Yale University, Electrical Engineering

PhD Research Assistant

2021–present

- Developed first-principles and many-body theory workflows for self-trapped excitons and excitonic polarons in real materials, connecting DFT, DFPT, GW/BSE-style methods, and reduced effective models.
- Built distributed simulation software in Python and C++ for leadership-class HPC systems, using MPI, OpenMP, CUDA, ROCm, PETSc, and SLEPc on Frontier, Perlmutter, and other national-lab clusters.
- Formulated coherent-state and path-integral viewpoints for exciton-phonon dynamics and used PyTorch, PyTorch Geometric, and equivariant graph neural networks to accelerate self-trapped-exciton calculations and related materials modeling workflows.
- Packaged reproducible scientific software in custom Docker containers and debugged distributed Linux applications with Bash and command-line tooling across HPC environments.
- Co-authored a 2025 *Journal of the American Chemical Society* paper on ultrafast self-trapped exciton formation in lead-free double halide perovskites and presented related APS research in 2024, 2025, and 2026.

Industry Experience

Probe Technologies Inc. / Weatherford Wireline Services

Software and Embedded Systems Engineer

2019–2021

- Developed CUDA-based software for high-volume acoustic waveform processing, visualization, and field-tool diagnostics in geological well logging systems.
- Built GUI applications with C#, ASP.NET server-side services, and Microsoft SQL Server workflows to automate remote calibration of temperature and pressure sensors in real time, improving production efficiency.
- Developed low-level debugging tools and engineering software for deployed instrumentation and server-side programs in hardware-adjacent production environments.

FindLight

Photonics Technical Writer

2018–2019

- Wrote clear technical content on lasers, optics, and photonics devices for engineering audiences and product-focused readers.

Selected Projects

Exciton-Phonon HPC Research: Developed distributed first-principles calculations for self-trapped excitons, many-body theory workflows, and scalable materials simulations.

Equivariant Scientific ML: Built PyTorch and PyTorch Geometric workflows with equivariant graph neural networks to accelerate molecular and excitonic modeling tasks.

Meep Ring Resonator FDTD: Designed a nanophotonic ring-resonator band-pass filter in telecom bands using Meep-based FDTD simulation workflows.

FEniCS-Based FEA: Built FEniCS-based finite-element workflows for frequency analysis of steel pipes in well environments to support physics-driven engineering studies.

Education

Yale University

PhD, Electrical Engineering

2021–present

Yale University

MPhil, Electrical Engineering

2023

Yale University

MS, Electrical Engineering

2023

Cornell University

MEng, Electrical and Computer Engineering

2017–2018

Cornell University

BS, Electrical and Computer Engineering

2013–2017

Technical Skills

Programming: Python, C, C++, CUDA, JavaScript, Bash, SQL, C#

Scientific Computing: MPI, OpenMP, PETSc, SLEPc, NumPy, SciPy, SymPy, matplotlib, FEniCS

Machine Learning: PyTorch, PyTorch Geometric, scikit-learn, equivariant graph neural networks

Platforms: Linux, macOS, Windows, Docker, Docker Compose, Kubernetes, gcloud, Frontier, Perlmutter

Domains: First-principles materials modeling, many-body theory, distributed computing, computational physics, numerical PDEs